Learning Domain-Specific Dimensionality Reduction: A Dataset Generation Pipeline

P. Gerdes[1]

[1]Film University Babelsberg KONRAD WOLF, Germany

Abstract
To-Do: Abstract

1. Introduction

Dimensionality reduction techniques such as PCA, t-SNE, and UMAP are widely used to visualize high-dimensional data, enabling intuitive exploration of complex datasets [STN*16]. However, these methods typically require access to the entire dataset in advance, limiting their applicability in dynamic environments where new data is continuously added. This paper addresses part of the challenge of approximating such dimensionality reduction methods for specific data domains using predictive modeling.

We/I (or passive?) propose an approach that learns to predict low-dimensional coordinates for raw token inputs by leveraging semantic embeddings followed by traditional dimensionality reduction. This supervised learning approach reduces computational overhead and supports incremental data addition. As a first step towards training such a model, this paper incorporates the following contributions:

- A comparison of selected dimensionality reduction techniques commonly used for visualizing text data, along with a brief analysis of their strengths and limitations in generating interpretable 2D embeddings (Section 3.1)
- A reproducible pipeline for constructing datasets that map text tokens to 2D coordinates using semantic embeddings and traditional dimensionality reduction methods (Section 3.2)
- An evaluation of the generated dataset using common dataset quality metrics (Section 3.4)

2. Related Work

To-Do: Related Work

3. Dataset Generation for Dimensionality Reduction

To-Do: Main Section

3.1. Comparison

To-Do: Comparison

3.2. Pipeline

To-Do: Pipeline

3.3. Results

To-Do: Results

3.4. Evaluation

To-Do: Evaluation

3.5. Discussion

To-Do: Discussion

4. Conclusion

To-Do: Conclusion

References

[STN*16] SMILKOV D., THORAT N., NICHOLSON C., REIF E., VIÉGAS F. B., WATTENBERG M.: Embedding Projector: Interactive Visualization and Interpretation of Embeddings, Nov. 2016. arXiv:1611.05469, doi:10.48550/arXiv.1611.05469. 1